

# Benjamin Dean *Senior Software Engineer*

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📍 Atlanta, GA

🌐 Linedin

🔗 Git

## Summary

Senior Software Engineer with 10+ years of experience designing large-scale frontend infrastructure, high-performance web applications, and real-time visualization platforms. Specialized in React, TypeScript, Python, and Go, with deep expertise in browser performance, streaming data architectures, and observability systems. Contributed to FullStory's core session replay platform serving hundreds of millions of user sessions annually, and currently leading frontend architecture for AI-powered Linux observability products leveraging eBPF and real-time telemetry. Experienced in driving architectural decisions, improving developer productivity, and delivering scalable user experiences for complex engineering platforms.

## Experience

### Senior Software Engineer

01/2024 – Present | Atlanta, GA

#### *Trench*

- Lead frontend architecture for an AI-native Linux observability platform built around eBPF-powered telemetry and real-time infrastructure monitoring.
- Architected a high-performance canvas rendering engine capable of visualizing dense streaming telemetry with low-latency updates and smooth user interactions.
- Designed a modular visualization framework supporting interactive timelines, event graphs, analytics dashboards, and system dependency exploration.
- Built frontend integration layers connecting continuously streaming backend pipelines with responsive React applications while maintaining efficient state synchronization.
- Optimized rendering performance for high-frequency event streams through update batching, render profiling, and memory optimization techniques.
- Introduced reusable React design patterns that standardized analytics interfaces and accelerated feature development across engineering teams.
- Integrated AI-assisted incident summarization capabilities that transform low-level telemetry into actionable operational insights for engineers.
- Partnered closely with backend, infrastructure, and product teams to define scalable data contracts between eBPF collectors, streaming services, and frontend applications.
- Established frontend performance profiling standards covering render tracing, memory analysis, and continuous optimization for real-time workloads.

### Tech Stack:

React • TypeScript • Go • Python • Canvas • eBPF • Streaming Systems • Linux • Performance Engineering

**Senior Software Engineer: Playback**

02/2022 – 12/2023

*FullStory*

Atlanta, Georgia, United States

- Led architecture for FullStory's next-generation session replay platform, establishing long-term technical direction for the Playback team.
- Joined as the team's first engineer and helped define foundational architecture, engineering standards, and core technical decisions.
- Modernized legacy frontend infrastructure by migrating internal frameworks to React, improving maintainability and developer velocity.
- Unified fragmented replay timing logic into a scalable architecture supporting future platform evolution.
- Designed streaming data structures enabling efficient real-time playback and analysis of recorded user sessions.
- Collaborated across engineering organizations to align replay architecture with future platform capabilities.

**Tech Stack:**

React • TypeScript • JavaScript • Streaming Data • Performance Engineering

**Senior Software Engineer (SWE III): Web Capture**

03/2019 – 02/2022

*FullStory*

Atlanta, Georgia, United States

- Delivered enterprise capabilities for detecting user frustration signals in real time, expanding behavioral analytics functionality for enterprise customers.
- Developed interactive tooling allowing customers to preview production configuration changes before deployment.
- Collaborated across multiple engineering teams to deliver customer-facing analytics and configuration management features.
- Built browser performance utilities that nearly eliminated forced style recomputation overhead across critical rendering paths.
- Created a JavaScript microbenchmarking framework that improved performance validation and optimization workflows for frontend engineers.

**Tech stack:**

JavaScript • React • Browser APIs • Performance Optimization • Enterprise Web Applications

**Software Engineer: Web Capture**

10/2016 – 03/2019 | Atlanta, GA

*FullStory*

- Contributed to FullStory's browser capture platform responsible for recording hundreds of millions of user sessions annually.
- Expanded compatibility for modern frontend ecosystems by adding support for CSS-in-JS frameworks within session replay infrastructure.

- Designed efficient client-side resource upload mechanisms that improved reliability and scalability of browser capture systems.
- Implemented GDPR-compliant privacy controls allowing applications to dynamically adjust data collection based on user consent.
- Produced technical documentation and customer-facing reports supporting enterprise deployments and compliance initiatives.

#### **Tech Stack:**

JavaScript • Browser APIs • Session Replay • Privacy Engineering • GDPR

#### **Software Engineering Intern**

05/2015 – 08/2015

*Google*

Mountain View, CA

- Developed frontend features for Google+ using Google Closure and Guice.
- Implemented content sharing capabilities for Google's next-generation web experience.
- Built image editing tools allowing users to crop and customize uploaded photos.
- Designed asynchronous JavaScript utilities that simplified client-side synchronization.

#### **Software Engineering Intern**

05/2014 – 08/2014

*Evernote*

Redwood City, CA

- Delivered UI and backend improvements to enhance content sharing workflows.
- Redesigned image gallery experience for user-generated content.
- Implemented legacy browser compatibility features for enterprise users.

#### **Software intern**

2012 – 2013

*LogicBlox*

- Developed and maintained Datalog-based test systems, identifying edge-case failures.
- Built OLAP data exploration frontend with tabular and graphical visualization components.

## **Education**

**Bachelor of Science (BS), Mathematics and  
Computer Science**

2012 – 2016

*Emory University*

## **Top Skills**

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### **Programming Languages**

Python, TypeScript, Go, JavaScript

### **Frontend Engineering**

React, Canvas-based UI systems, SPA architecture, real-time dashboards

### **Backend & Systems**

Distributed systems, streaming data pipelines, event-driven architecture, REST APIs

### **Performance Engineering**

Web performance optimization, rendering optimization, memory profiling, microbenchmarking

### **Observability & Analytics**

Session replay systems, user behavior analytics, system telemetry, eBPF-based observability

### **Tools & Practices**

Git, Agile development, debugging at scale, production monitoring, cross-team architecture design

### **AI / Applied AI**

AI-assisted analytics features, LLM-integrated UI workflows, data summarization systems